



Centre for
Robotics



Ensemble-based event camera place recognition under varying illumination

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Place Recognition under Varied Illumination

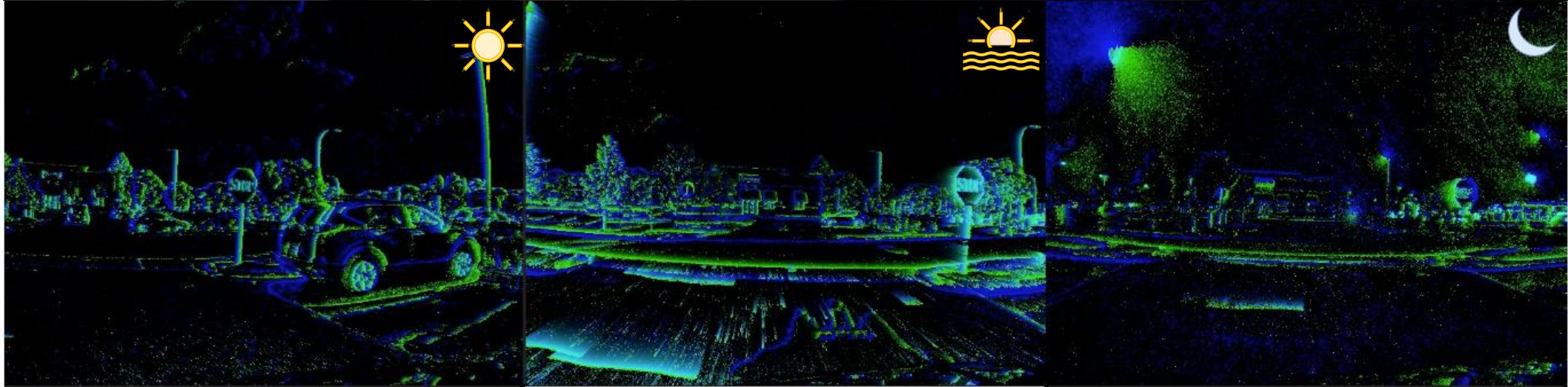


Places at various times of day

Conventional Camera:

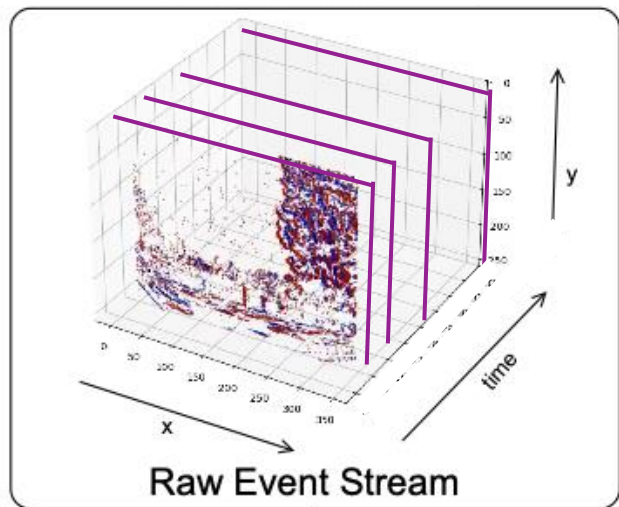


Event Camera:



Method: Generating place descriptors

A. Raw events sliced into 4 bins per location



*Each bin has varying resolution
(1, 0.5, 0.25, 0.1 secs)*

B. 4 Event to image reconstruction

Event Count

Event Count (polarity summed)

Time Surface [1]

E2VID [2]

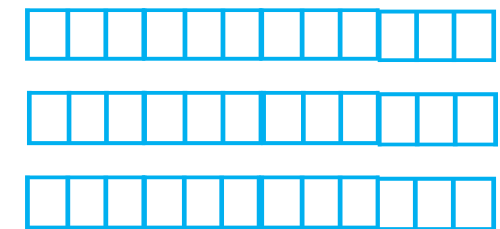
C. 4 SoTA VPR Feature extractors

Megaloc [3]

CosPlace [4]

MixVPR [5]

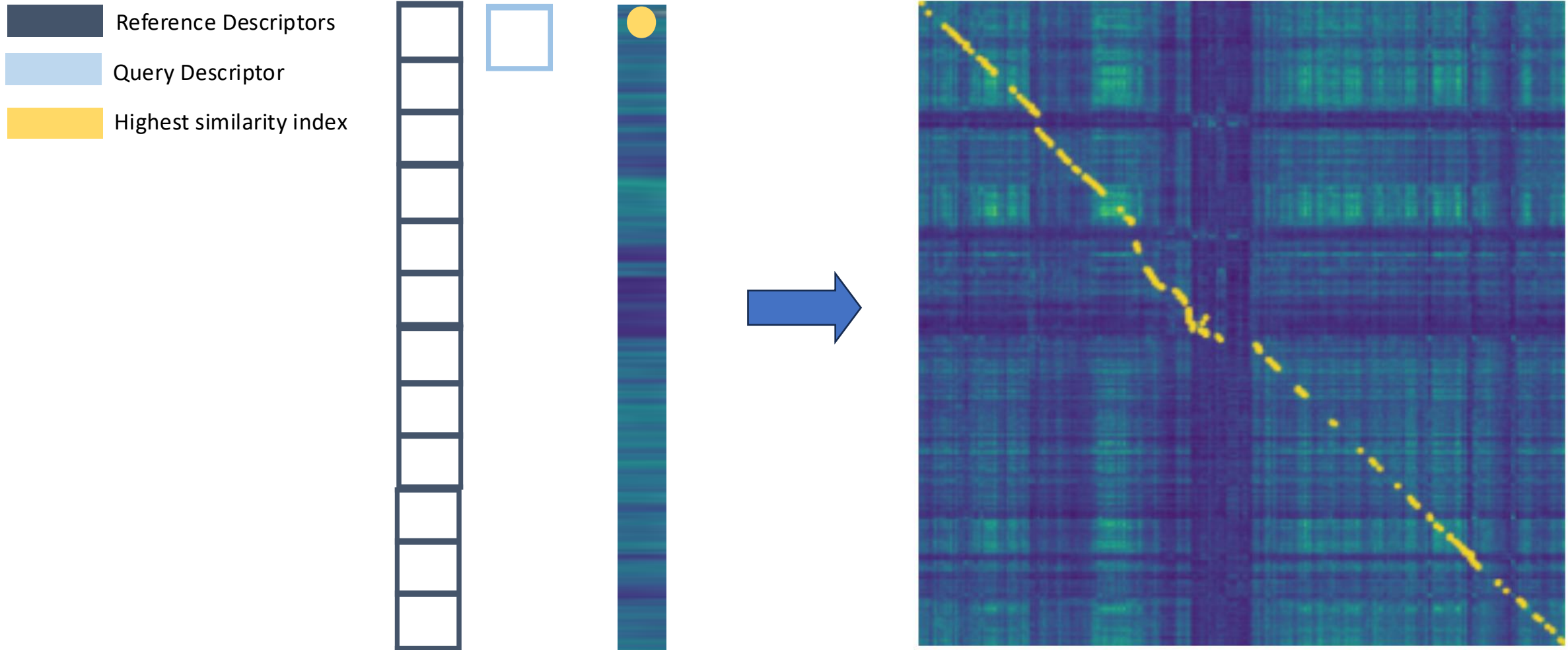
NetVLAD [6]



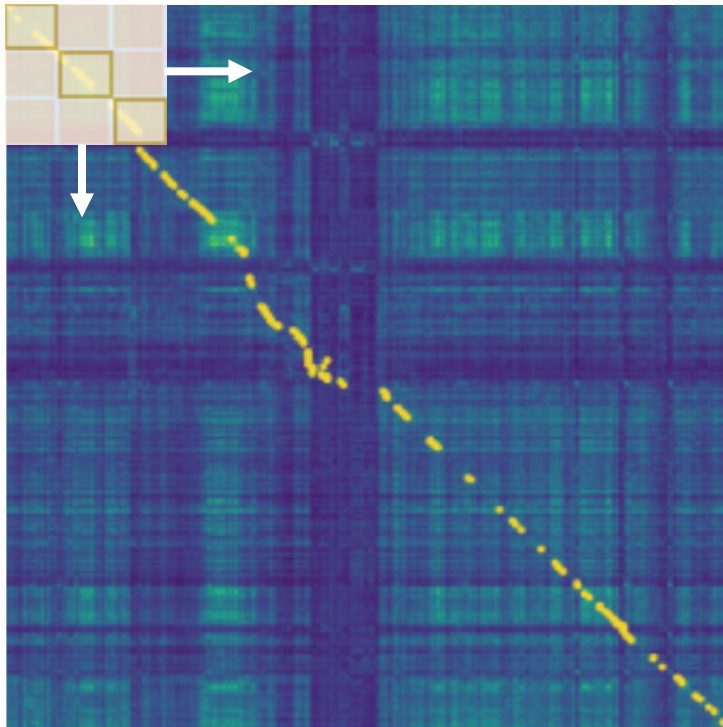
*64 combinations of
descriptors per location*

- [1] Sironi et al., CVPR 2018 (Time Surface)
- [2] Rebecq et al., TPAMI 2021 (E2VID)
- [3] Berton & Masone, CVPR Workshops 2025 (MegaLoc)
- [4] Berton et al., CVPR 2022 (CosPlace)
- [5] Ali-bey et al., CVPR 2023 (MixVPR)
- [6] Arandjelović et al., CVPR 2016 (NetVLAD)

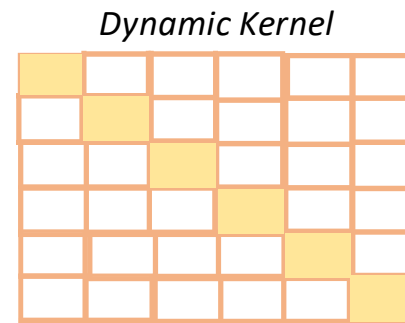
Similarity Matrix for Place Recognition



Method: Modified sequence matching



Convolution based sequence matching [1,2]



Row and Column Normalization

$$Z = \frac{x - \mu}{\sigma}$$



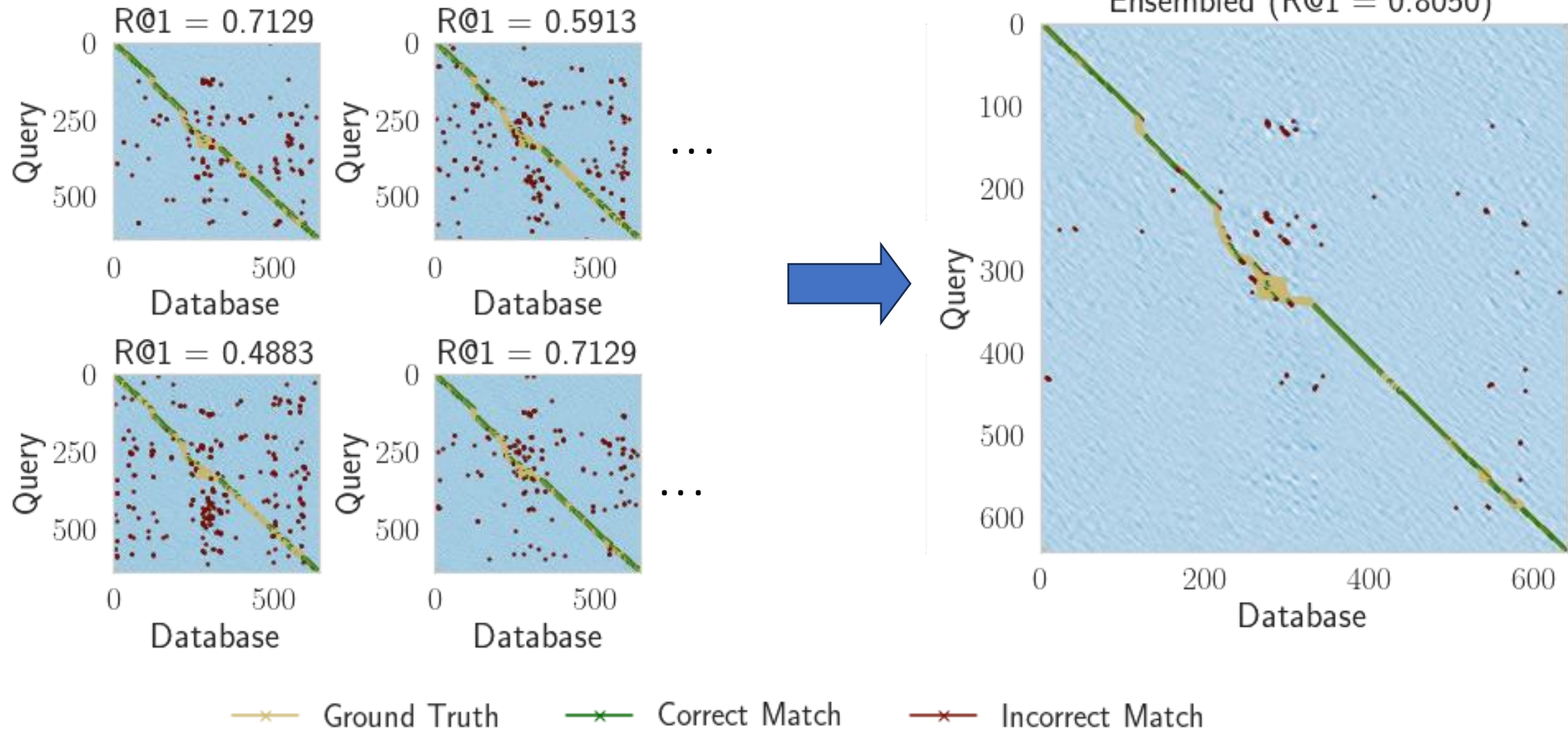
MEAN RECALL@1 ACROSS SEQUENCE LENGTHS

Seq Len	SeqSLAM	Ours	p-value
10	0.3473	0.3410	0.6874
20	0.4301	0.4613	0.0554
30	0.4593	0.4996	0.0096

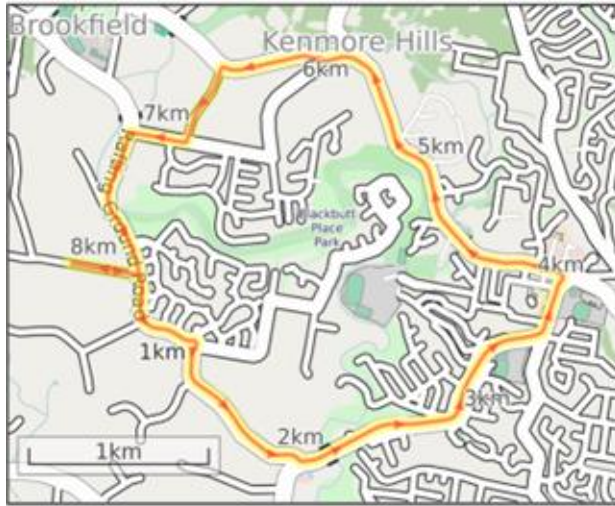
[1] Milford & Wyeth, ICRA 2012 (SeqSLAM)

[2] Garg et al., CORL 2022 (SeqMatchNet)

Method: Aggregating predictions

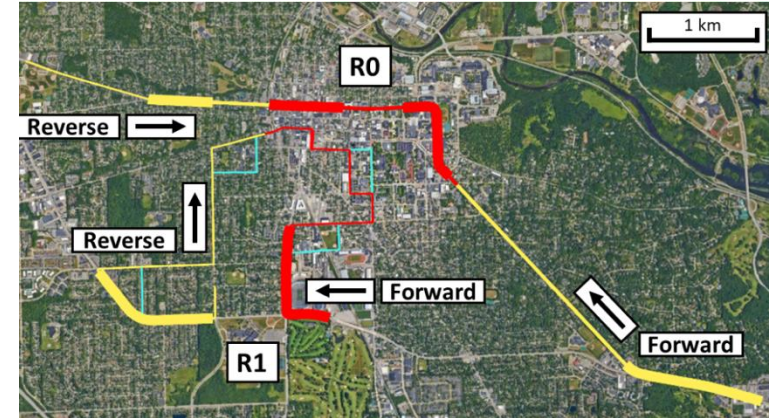


Datasets evaluated



Brisbane Event VPR Dataset [1]

- 8km route across sunset, morning, sunrise, day, night
- We test 1 reference across 5 query traverses
- DAVIS 346 event camera



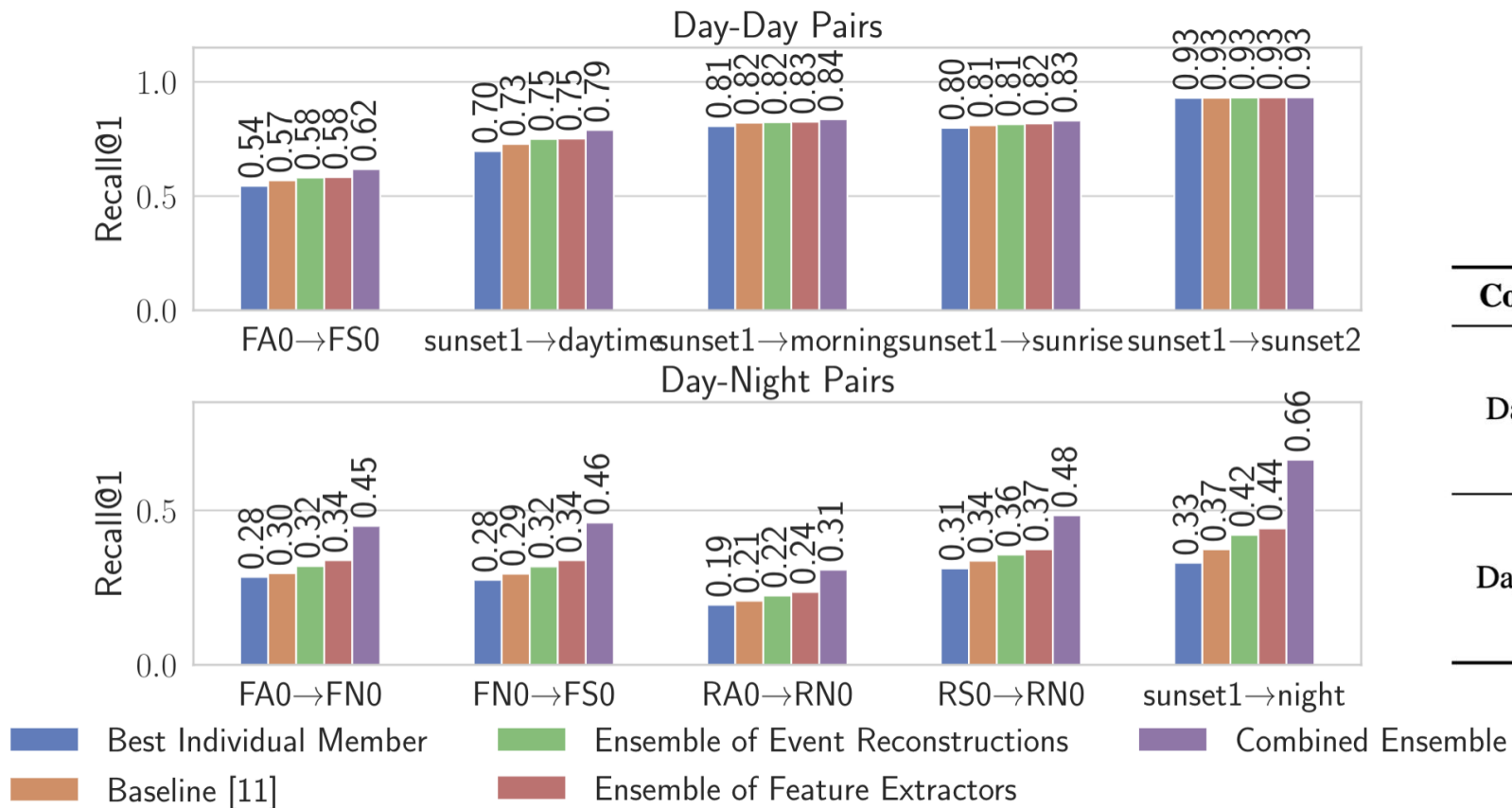
Novel Sensors for Autonomous Vehicle Perception Dataset (NSAVP) [2]

- 8.3km route with forward and reverse traverses across afternoon, sunset and night
- We test 5 reference across 3 query traverses
- Uses Prophesee Gen4 event sensor

[1] Fischer & Milford, IEEE RA-L 2020 (Brisbane-Event-VPR dataset)

[2] Carmichael et al., IJRR 2024 (NSAVP dataset)

Results for day-night performance gain

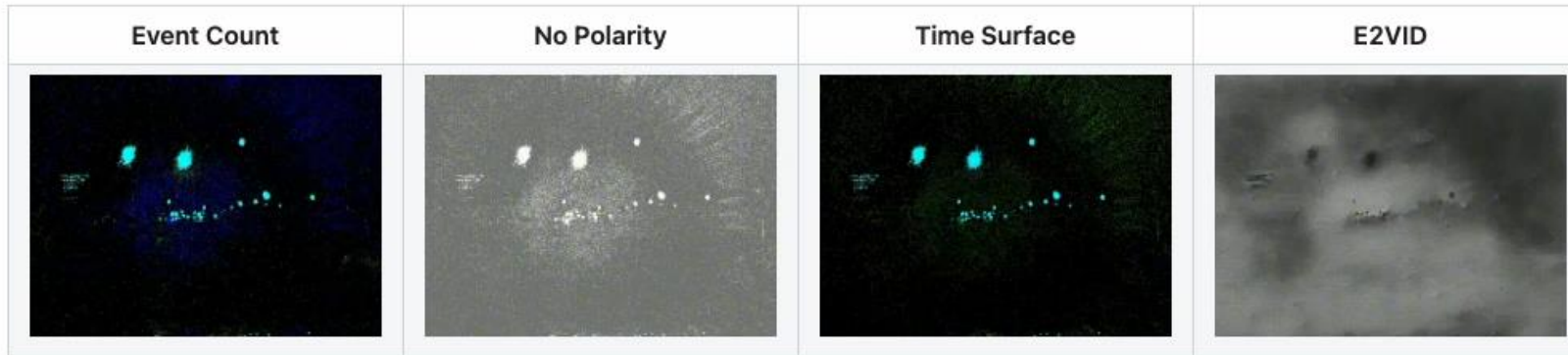


Performance Improvement over baseline (ensemble of temporal resolution)

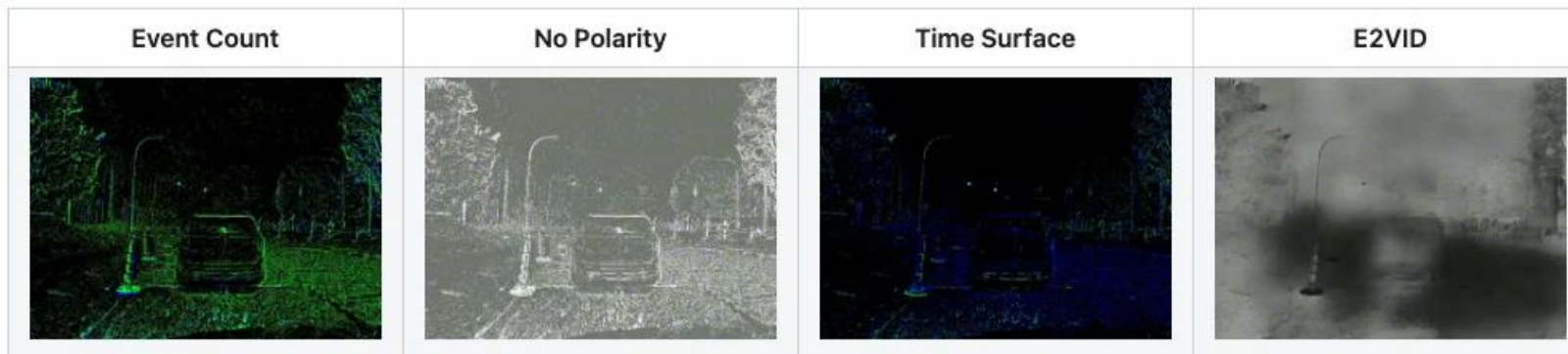
Condition	Method	R@1 (%)	Gain (%)
Day-Day	Baseline [11]	77.16	—
	Reconstruction	78.00	1.09
	Feature Extractor	78.19	1.33
	Combined Ensemble	80.13	3.85
Day-Night	Baseline [11]	30.20	—
	Reconstruction	32.79	8.58
	Feature Extractor	34.58	14.51
	Combined Ensemble	47.31	56.86

Results across event reconstruction methods

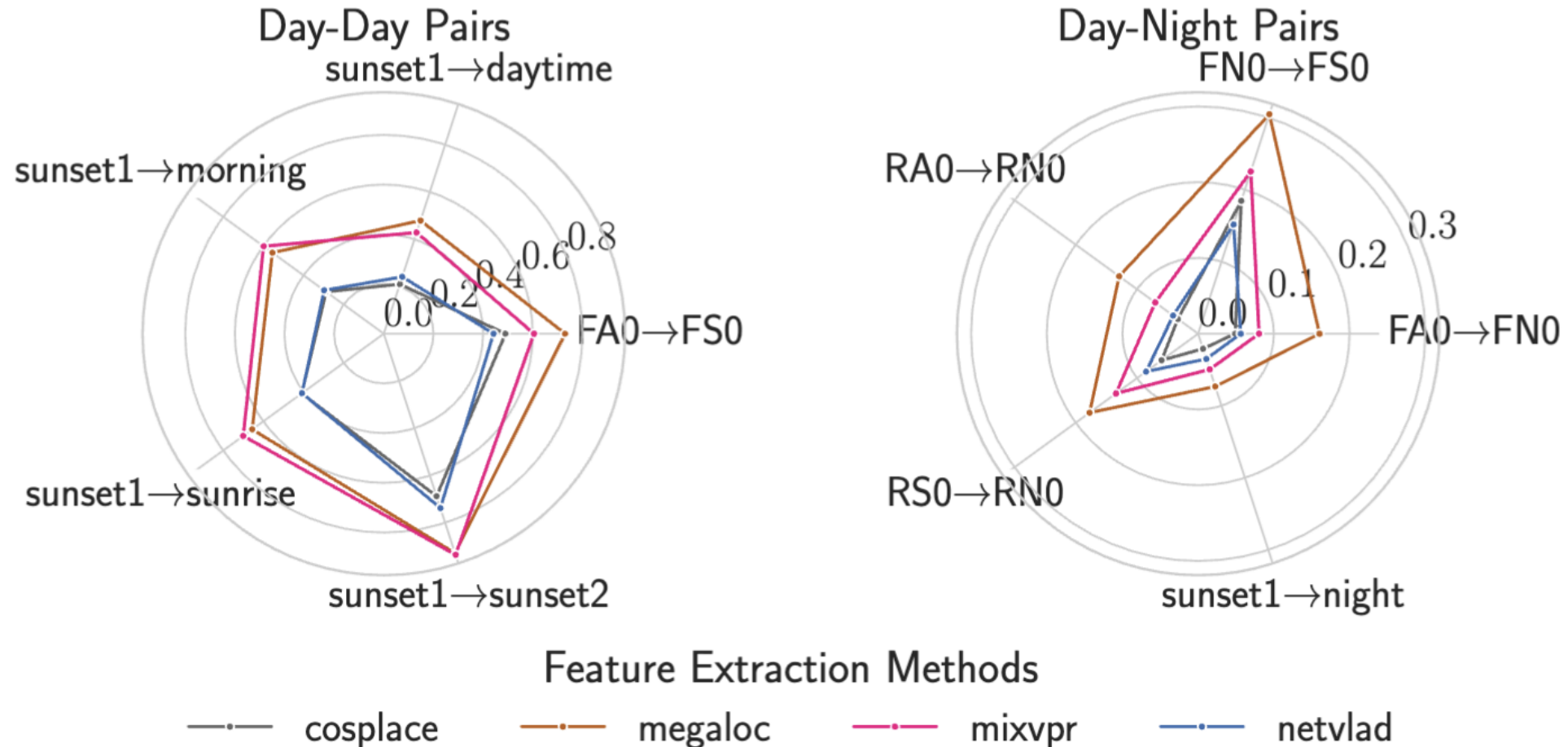
NSAVP night



NSAVP afternoon



Results across VPR feature extractors



Limitations

- Ensemble methods improve robustness but increase computational cost compared to the individual ensemble member
- None of the feature extractors used were trained on event data, limiting their ability to exploit the unique properties of asynchronous events.
- The approach was tested only in driving scenarios; performance in indoor or unstructured natural settings is unverified.